

SYDNEY GIRLS HIGH SCHOOL



YEAR 12 4 UNIT MATHEMATICS

March 1999

Time Allowed: 90 minutes.

Instructions:

- There are sixteen (16) questions.
- Attempt all questions
- Questions are not of equal value
- All necessary working should be shown. Marks may be deducted for careless or badly arranged work.
- Write on one side of the paper only

Name: _____

Class: _____

Marks

Question 1

8

If $z = \sqrt{3} + i$

- a) find
- $\arg z$
 - $|z|$
 - $\bar{z}$
- b) show on the Argand Diagram
- z
 - $\bar{z}$
 - iz
 - $\frac{1}{z}$
 - $z + iz$

Question 2

4

If $z = 4 - 3i$

- a) Show that
- $z\bar{z}$ is real
 - $z + \bar{z}$ is real
- b) Find a and b if $\frac{1}{z} = a + ib$

Question 3

4

A represents the complex number $z = 2 + 2i$,

B represents a 60° anticlockwise rotation of A, and,

C represents a 90° clockwise rotation of A.

Write down the complex numbers represented by B and C

Question 4

3

If $z = -\sqrt{3} + i$

- express z in mod-arg form
- express z^6 in the form $a + ib$

Question 5

6

Sketch the region common to

- $1 \leq R(z) \leq 3$ and $\frac{\pi}{6} \leq \arg z \leq \frac{\pi}{3}$
- $0 \leq \arg(z+4) \leq \frac{2\pi}{3}$ and $|z+4| \leq 4$

Question 6

6

- a) Express $\sqrt{24 - 70i}$ in the form $a + ib$, where a and b are real and $a > 0$.
- b) Hence, or otherwise, solve $z^2 - (1-i)z - 6 + 17i = 0$

Question 7

7

- a)
- Find the locus of z if $|z - 4i| = 2|z + 1|$
 - Describe this locus.
- b) Determine the locus of w , if $|z| = 2$ and $w = \frac{z-1}{z}$

Question 8

4

Given $|z+1|=1$, find the maximum and minimum of $\arg(z+3)$

Question 9

8

Find the locus of z if $w = \frac{z-2i}{z+2}$ is

- i) purely real
- ii) purely imaginary

Sketch the locus in each case.

Question 10

4

Find and sketch the locus of z where $\arg\left(\frac{-z}{i}\right) = \arg\left(\frac{1}{z}\right)$

Question 11

6

Given $z = \cos\theta + i\sin\theta$, express $\cos^4\theta$ in terms of $\cos 4\theta$ and $\cos 2\theta$.

Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \cos^4\theta \, d\theta$

Question 12

4

The points A, B, C, and D are the vertices of the quadrilateral ABCD. They represent the complex numbers z_1, z_2, z_3, z_4 respectively such that

$$\arg(z_1 - z_4) = \arg(z_2 - z_3) \text{ and } |z_4 - z_3| = |z_1 - z_2|.$$

Describe the quadrilateral ABCD, including a specific name, justifying your answer with appropriate reasons and illustrations.

Question 13

10

Find the seven seventh roots of 1, ie solve $z^7 - 1 = 0$

- i) Plot the roots on the Argand Diagram.
- ii) If $w = \text{cis} \frac{2\pi}{7}$, show that $1 + w + w^2 + w^3 + w^4 + w^5 + w^6 = 0$
- iii) Factorise $z^7 - 1$ into real quadratic factors.

Question 14

8

- a) If the polynomial $P(x)$ has a root at $x = a$ of multiplicity m , show that the derived polynomial $P'(x)$ has a root at $x = a$ of multiplicity $m-1$.
- b) A polynomial $P(x) = 8x^4 - 20x^3 - 18x^2 + 81x - 54$ is known to have a triple root. Find the roots of this equation and sketch the curve, showing the essential features.

Question 15

6

Given that $(2-i)$ is a zero of $2x^3 + mx^2 + nx + 15$, determine m and n , where m and n are real. Hence, factorise $2x^3 + mx^2 + nx + 15$ over the real field.

Question 16

12

Given α, β, γ are the roots of $x^3 + 2x^2 - 3x + 4 = 0$

- a) find
 - i) $\alpha^2 + \beta^2 + \gamma^2$
 - ii) $\alpha^3 + \beta^3 + \gamma^3$
 - iii) $\alpha^4 + \beta^4 + \gamma^4$
- b) find the equation whose roots are
 - i) $\alpha + 2, \beta + 2, \gamma + 2$
 - ii) $\alpha^2, \beta^2, \gamma^2$
 - iii) $\frac{\alpha\beta}{\gamma}, \frac{\alpha\gamma}{\beta}, \frac{\beta\gamma}{\alpha}$

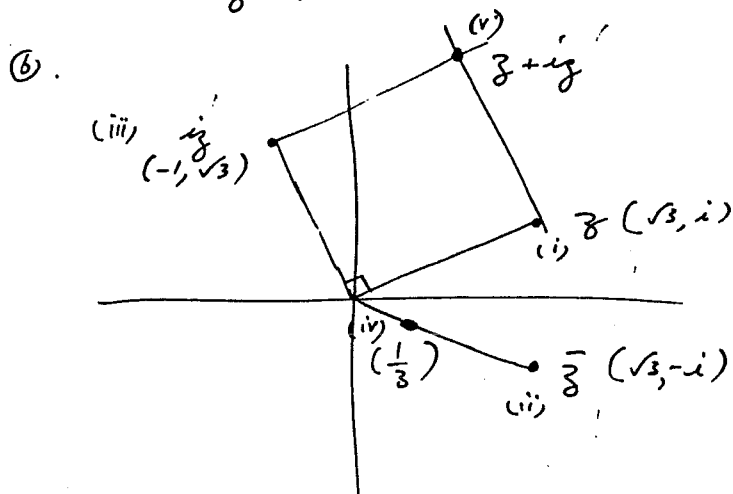
End of Exam

① $z = \sqrt{3} + i$

② (i) $\tan \theta = \frac{1}{\sqrt{3}} = \frac{\pi}{6} \therefore \arg z = \frac{\pi}{6}$

(ii) $|z| = \sqrt{x^2 + y^2} = \sqrt{(\sqrt{3})^2 + 1^2} = \underline{2}$

(iii) $\bar{z} = \sqrt{3} - i$



$\frac{1}{z} = \frac{1}{\sqrt{3} + i} \cdot \frac{\sqrt{3} - i}{\sqrt{3} - i} = \frac{1}{4} (\sqrt{3} - i)$

②. $z = 4 - 3i$

(i) $z \cdot \bar{z} = (4 - 3i)(4 + 3i) = 16 + 9 = 25$
 $\therefore \text{Real}$

(ii) $z + \bar{z} = (4 - 3i) + (4 + 3i) = 8$
 $\therefore \text{Real}$

$\frac{1}{z} = a + ib = \frac{1}{4 - 3i} \cdot \frac{4 + 3i}{4 + 3i}$
 $= \frac{4 + 3i}{25}$

$\therefore a = \frac{4}{25} \quad b = \frac{3}{25}$

③ $z = 2 + 2i$

60° rotation anticlockwise

is $(\frac{1}{2} + \frac{\sqrt{3}}{2}i)$

$\therefore$ Co-ords of B. $(2 + 2i)(\frac{1}{2} + \frac{\sqrt{3}}{2}i)$

$z_1 = (1 + \sqrt{3}) + i(1 + \sqrt{3})$ 2

90° rotation clockwise is $-i$

$\therefore z_2 = 2 - 2i$

Co-ords of C $(2, -2i)$ 2

④ (i) $z = -\sqrt{3} + i$ $\tan \theta = \frac{1}{-\sqrt{3}}$

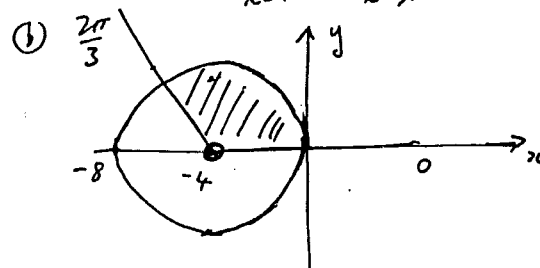
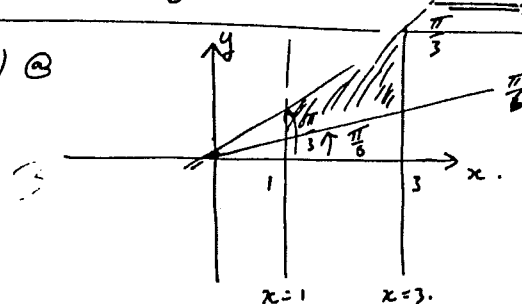
$|z| = \sqrt{(\sqrt{3})^2 + 1^2} = 2 \therefore \arg z = \frac{5\pi}{6}$

$\therefore z = 2 \operatorname{cis} \frac{5\pi}{6}$

(ii) Now $z^6 = (2 \operatorname{cis} \frac{5\pi}{6})^6 = 2^6 \operatorname{cis} 5\pi$ (By De Moivre's)
Then

$z^6 = 64 \operatorname{cis} \pi = \underline{\underline{-64 + 0i = -64}}$

⑤ ②



3

⑥ (i) $\sqrt{24-70i} = a+ib$

$$24-70i = a^2 - b^2 + 2abi$$

$$\therefore a^2 - b^2 = 24 \quad ab = -35$$

$$\therefore a = 7 \quad \text{or} \quad a = -7$$

$$b = -5 \quad \text{or} \quad b = 5$$

but $a > 0$.

$$\therefore \underline{7-5i}$$

3

(ii) $z^2 - (1-i)z - 6+17i = 0$

$$z = \frac{1-i \pm \sqrt{(1-i)^2 - 4(-6+17i)}}{2}$$

$$z = \frac{1-i \pm \sqrt{1-2i-1+24-68i}}{2} = \frac{1-i \pm \sqrt{24-70i}}{2}$$

$$\therefore z = \frac{(1-i) \pm (7-5i)}{2}$$

$$z = \underline{4+3i} \quad \text{or} \quad z = \underline{-3+2i}$$

Q7 a) $|z-4i| = 2|z+1|$

$$x^2 + (y-4)^2 = 2^2 \{ (x+1)^2 + y^2 \}$$

$$x^2 + y^2 - 8y + 16 = 4 \{ x^2 + 2x + 1 + y^2 \}$$

3 $0 = 3x^2 + 3y^2 + 8x + 8y - 12$

$$x^2 + y^2 + \frac{8}{3}x + \frac{8}{3}y - 4 = 0$$

$$(x + \frac{4}{3})^2 + (y + \frac{4}{3})^2 = 4 + \frac{16}{9} + \frac{16}{9} = \frac{68}{9}$$

Circle: Centre $(-\frac{4}{3}, -\frac{4}{3})$

Radius $\underline{\underline{\frac{\sqrt{68}}{3} \text{ units}}}$

Q7 b) $w = \frac{z-1}{z}$

$$wz = z - 1$$

$$1 = z - wz = z(1-w)$$

$$\therefore z = \frac{1}{1-w}$$

$$|z| = \left| \frac{1}{1-w} \right|$$

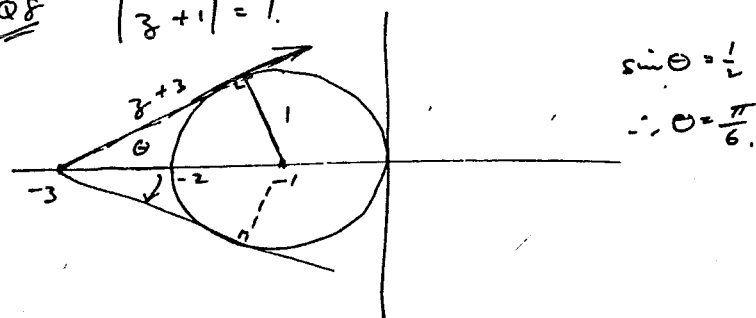
$$\therefore 2 = \left| \frac{1}{w-1} \right| \Rightarrow |w-1| = \frac{1}{2}$$

$\therefore$ Locus of w is a circle
centre $(1, 0)$ Radius $\frac{1}{2}$

3

Q8

$$|z+1| = 1$$



$$\sin \theta = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{6}$$

$$\therefore \text{Max arg is } \frac{\pi}{6}$$

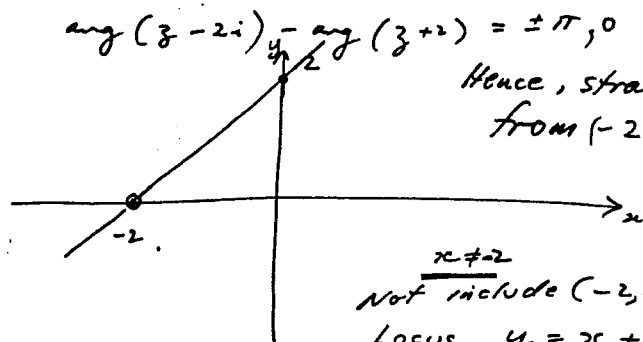
$$\text{Min arg is } -\frac{\pi}{6}$$

⑨. (i) $w = \frac{z-2i}{z+2}$ and w is purely real.

then: $\arg\left(\frac{z-2i}{z+2}\right) = \pm\pi, 0$

$\arg(z-2i) - \arg(z+2) = \pm\pi, 0$

Hence, straight line
from $(-2, 0)$ to $(0, 2)$



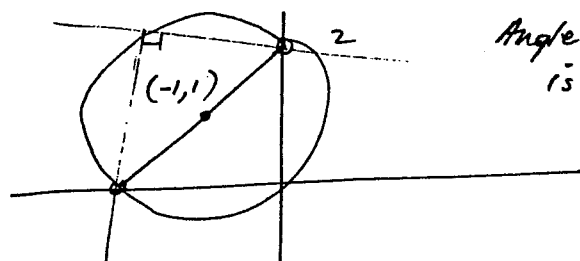
$x \neq -2$
Not include $(-2, 0)$.

Locus $y = x + 2$

(ii) $w = \frac{z-2i}{z+2}$ and w is purely imaginary.

then $\arg\left(\frac{z-2i}{z+2}\right) = \pm\frac{\pi}{2}$.

then $\arg(z-2i) - \arg(z+2) = \pm\frac{\pi}{2}$.



Angle in semi-circle
is a right angle:

Locus: Circle Centre $(-1, 1)$
Radius $\sqrt{2}$ Units

or $(x+1)^2 + (y-1)^2 = 2$

Q9

$w = \frac{z-2i}{z+2}$

$\therefore w = \frac{z-2i}{z+2} \times \frac{\bar{z}+2}{\bar{z}+2}$

$z = x + iy$

$\bar{z} = x - iy$

$w = \frac{z \cdot \bar{z} - 2i\bar{z} + 2z - 4i}{(z+2)(\bar{z}+2)}$

$w = \frac{x^2 + y^2 - 2i(x - iy) + 2(x + iy) - 4i}{(z+2)(\bar{z}+2)}$

$w = \frac{x^2 + y^2 - 2ix - 2y + 2x + 2iy - 4i}{(z+2)(\bar{z}+2)}$

(i) Now if w is purely real.

$-2ix + 2iy - 4i = 0$. Exclude $(-2, 0)$.

i.e., $2y = 2x + 4$
 $y = x + 2$

(ii) Now if w is purely imaginary.

$x^2 + y^2 + 2x - 2y = 0$.

$(x^2 + 2x + 1) + (y^2 - 2y + 1) = 2$.

$(x+1)^2 + (y-1)^2 = 2$.

$\therefore$ Circle Centre $(-1, 1)$
Radius $\sqrt{2}$.

Exclude $(-2, 0)$ and $(0, 2)$
(end points of diameter)

⑩. $\arg(-\frac{z}{i}) = \arg(\frac{1}{z})$.

$\therefore \arg(-z) - \arg(i) = \arg 1 - \arg z$.

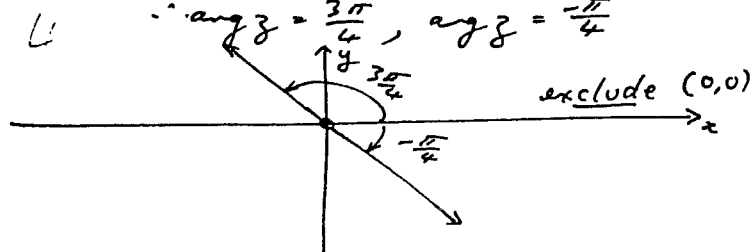
$\arg(-1 \cdot z) - \frac{\pi}{2} = 0 - \arg z$.

$\arg z + \arg(-1) + \arg z = \frac{\pi}{2}$.

$2\arg z + \pi = \frac{\pi}{2}$

$2\arg z = \frac{3\pi}{2} \text{ or } -\frac{\pi}{2}$

$\therefore \arg z = \frac{3\pi}{4}, \arg z = -\frac{\pi}{4}$



⑪ $z = \cos \theta + i \sin \theta$.

$z + \bar{z} = \cos \theta + i \sin \theta + \cos \theta - i \sin \theta$.

$\therefore z + \bar{z} = 2 \cos \theta \Rightarrow (\frac{1}{z} + z)$

$z^4 = (\cos \theta + i \sin \theta)^4$

$\therefore (2 \cos \theta)^4 = (z + \frac{1}{z})^4$

$= z^4 + \frac{1}{z^4} + 4z^3 \cdot \frac{1}{z} + 6z^2 \cdot \frac{1}{z^2} + 4z \cdot \frac{1}{z^3}$

$= z^4 + \frac{1}{z^4} + 4(z^2 + \frac{1}{z^2}) + 6$.

Now: $z^4 + \frac{1}{z^4} = 2 \cos 4\theta$.

$z^2 + \frac{1}{z^2} = 2 \cos 2\theta$.

$\therefore (2 \cos \theta)^4 = 2 \cos 4\theta + 8 \cos 2\theta + 6$.

$\therefore 16 \cos^4 \theta = 2 \cos 4\theta + 8 \cos 2\theta + 6$.

$\cos^4 \theta = \frac{\cos 4\theta + 4 \cos 2\theta + 3}{8}$

By De Moivre's Thm
 $z^n = \cos n\theta + i \sin n\theta$

$z^{-n} = \cos n\theta - i \sin n\theta$

$\therefore z^n + \frac{1}{z^n} = 2 \cos n\theta$.

⑫ $I = \int_0^{\frac{\pi}{2}} \cos^4 \theta \cdot d\theta$.

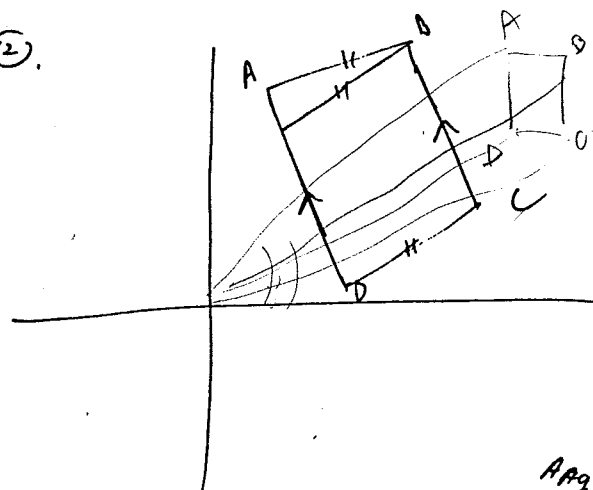
$I = \frac{1}{8} \int_0^{\frac{\pi}{2}} (\cos^4 \theta + 4 \cos 2\theta + 3) \cdot d\theta$.

$I = \frac{1}{8} \left[\frac{1}{4} \sin 4\theta + 4 \cdot \frac{1}{2} \sin 2\theta + 3\theta \right]_0^{\frac{\pi}{2}}$

$I = \frac{1}{8} \left[\frac{1}{4} \cdot 0 + 2 \cdot 0 + 3 \cdot \frac{\pi}{2} \right] - [0 + 0 + 0]$

$I = \frac{3\pi}{16}$

⑬



$\arg z_1 = \arg z_2$

$\arg(z_1 - z_4) = \arg(z_2 - z_3)$
means $AD \parallel BC$.

$|z_4 - z_3| = |z_1 - z_2|$
means $DC = AB$.

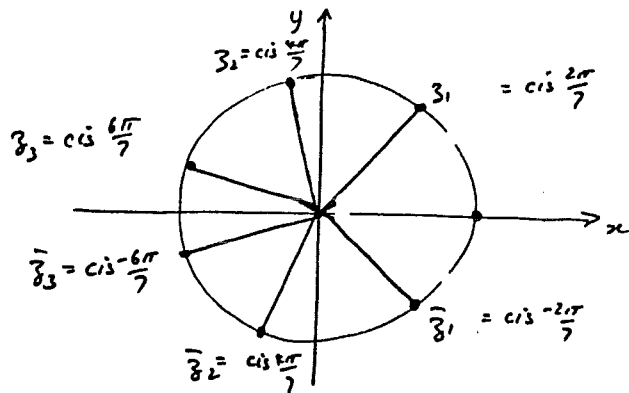
Now: $AD \parallel BC$
and $DC = AB$.

$\therefore ABCD$ is an isosceles trapezium
or a parallelogram.

(13) (i) $z^7 - 1 = 0$

Hence: $z^7 = 1 = \text{cis } 0$ let $z = \cos \theta + i \sin \theta$
 $z^7 = (\cos \theta + i \sin \theta)^7 = \text{cis } 7\theta$
 then $z^7 = \text{cis}(0 \pm 24\pi)$
 $\therefore z = \text{cis}\left(\frac{0 \pm 24\pi}{7}\right)$ by De Moivre's Thm
 and $k = 0, 1, 2, 3, 4, 5, 6$.

$\therefore z = \text{cis } 0, \text{cis } \frac{2\pi}{7}, \text{cis } \frac{4\pi}{7}, \text{cis } \frac{6\pi}{7}$
 $\text{cis } -\frac{2\pi}{7}, \text{cis } -\frac{4\pi}{7}, \text{cis } -\frac{6\pi}{7}$



(ii) Now let $w = \text{cis } \frac{2\pi}{7}$

then $w^2 = (\text{cis } \frac{2\pi}{7})^2 = \text{cis } \frac{4\pi}{7}$ by De Moivre's Thm.
 Hence $w^3 = \text{cis } \frac{6\pi}{7}, w^4 = \text{cis } \frac{8\pi}{7}, w^5 = \text{cis } \frac{10\pi}{7}, w^6 = \text{cis } \frac{12\pi}{7}, w^7 = \text{cis } 2\pi = 1$.

Now $z^7 = 1 \Rightarrow 1 - z^7 = 0$ and $z^7 = w^7$ when $z = w$.

$\therefore 1 - w^7 = (1 - w)(1 + w + w^2 + w^3 + w^4 + w^5 + w^6)$

But $(1 - w) \neq 0$ as $w \neq 1$

$\therefore (1 + w + w^2 + w^3 + w^4 + w^5 + w^6) = 0$.

OR $1 + w + w^2 + w^3 + w^4 + w^5 + w^6$ is a G.P. with $a = 1$
 $r = w$.

Now $S_n = \frac{a(r^n - 1)}{r - 1} \Rightarrow S_7 = \frac{1 \cdot (w^7 - 1)}{w - 1}$ but $w^7 = 1$.

$\therefore S_7 = \frac{0}{w - 1} = 0$.

$\therefore 1 + w + w^2 + w^3 + w^4 + w^5 + w^6 = 0$

(iv) Now $z^7 - 1 = (z - 1)(z - w)(z - w^2)(z - w^3)(z - w^4)(z - w^5)(z - w^6)$
 $= (z - 1)(z - w)(z - w^2)(z - w^3)(z - w^4)(z - w^5)(z - w^6)$

But w and w^6, w^2 and w^5, w^3 and w^4
 are conjugate pairs.

$z^7 - 1 = (z - 1)(z - w)(z - \bar{w})(z - w^2)(z - \bar{w}^2)(z - w^3)(z - \bar{w}^3)$

Now $w + \bar{w} = 2 \cos \frac{2\pi}{7}$

and $w \cdot \bar{w} = +1$.

Hence, by applying conjugate pairs.

$z^7 - 1 = (z - 1)(z^2 - 2 \cos \frac{2\pi}{7} z + 1)(z^2 - 2 \cos \frac{4\pi}{7} z + 1)(z^2 - 2 \cos \frac{6\pi}{7} z + 1)$

(14). (i) $\therefore P(x) = (x-a)^m \cdot Q(x)$

$$\frac{dP}{dx} = m \cdot (x-a)^{m-1} \cdot Q(x) + (x-a)^m \cdot Q'(x).$$

$$P'(x) = (x-a)^{m-1} \{ m \cdot Q(x) + (x-a) \cdot Q'(x) \}.$$

$$P'(x) = (x-a)^{m-1} \{ S(x) \}.$$

$\therefore$ If $x=a$ is a root of multiplicity m of $P(x)$
 $\therefore$ then $x=a$ is a root of multiplicity $(m-1)$ of $P'(x)$.

(ii) $P(x) = 8x^4 - 20x^3 - 18x^2 + 81x - 54$ (triple root)

$P'(x) = 32x^3 - 60x^2 - 36x + 81$ (double root)

$P''(x) = 96x^2 - 120x - 36$ (single root)

$P''(x) = 0 \Rightarrow 12(8x^2 - 10x + 3) = 0.$

$\therefore (4x+1)(2x-3) = 0.$

Now $x = \frac{3}{2}$ or $x = -\frac{1}{4}$
 but $x = -\frac{1}{4}$ does not satisfy $P(x)$.

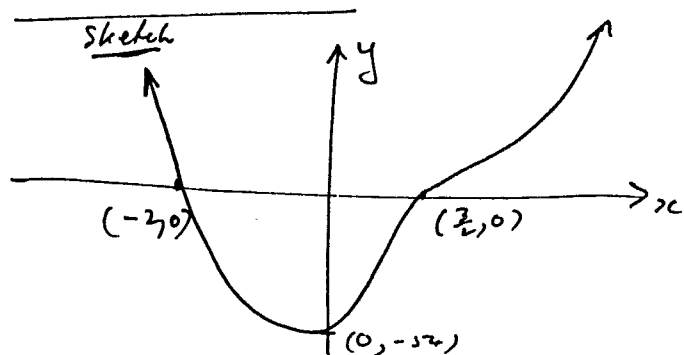
$\therefore$ triple root at $x = \frac{3}{2}$

$\therefore$ triple factor $(2x-3)$.

Now $\frac{-54}{8} = \left(\frac{3}{2}\right)^3 \cdot d.$

$\frac{-54}{8} = \frac{27}{8} \cdot d \Rightarrow d = -2.$

$\therefore P(x) = (2x-3)^3(x+2)$



Q15

$P(x) = 2x^3 + mx^2 + nx + 15$

and $x = 2-i$ is a zero

then $x = 2+i$ is a zero by the law of conjugate pairs.

Hence: $x^2 - 4x + 5$ is a factor.

Now: $x^2 - 4x + 5 \overline{) 2x^3 + mx^2 + nx + 15}$

$$\begin{array}{r} 2x+3 \\ 2x^3 + mx^2 + nx + 15 \\ \underline{2x^3 - 8x^2 + 10x} \\ (m+8)x^2 + (n-10)x + 15 \\ \underline{k(x^2 - 4x + 5)} \\ 0 \end{array}$$

$\therefore 5k = 15 \Rightarrow k = 3.$

$\therefore m+8-3 = 0 \quad n-10+12 = 0.$
 $m = -5 \quad n = -2.$

$\therefore P(x) = 2x^3 - 5x^2 - 2x + 15$

$P(x) = (2x+3)(x^2-4x+5)$

Q16 (a) $x^3 + 2x^2 - 3x + 4 = 0$.

(i) $\alpha^2 + \beta^2 + \gamma^2$
 $= (\alpha + \beta + \gamma)^2 - 2\alpha\beta$
 $= (-2)^2 - 2(-3) = 10$

$$\left. \begin{aligned} \alpha + \beta + \gamma &= \frac{-2}{1} = -2 \\ \alpha\beta + \beta\gamma + \gamma\alpha &= \frac{-3}{1} = -3 \\ \alpha\beta\gamma &= \frac{-4}{1} = -4 \end{aligned} \right\}$$

(ii) $\alpha^3 + \beta^3 + \gamma^3$ Now: $x^3 = -2x^2 + 3x - 4$
 $\therefore \alpha^3 = -2\alpha^2 + 3\alpha - 4$
 $\therefore \alpha^3 + \beta^3 + \gamma^3 = -2\alpha^2 + 3\alpha - 4 + 3\beta^2 - 4\beta + 3\gamma^2 - 4\gamma$
 $= -2(10) + 3(-2) - 12$
 $= -38$

(iii) $\alpha^4 + \beta^4 + \gamma^4$ Now $x^4 = -2x^3 + 3x^2 - 4x$
 $\therefore \alpha^4 = -2\alpha^3 + 3\alpha^2 - 4\alpha$
 $\therefore \alpha^4 + \beta^4 + \gamma^4 = -2\alpha^3 + 3\alpha^2 - 4\alpha - 2\beta^3 + 3\beta^2 - 4\beta - 2\gamma^3 + 3\gamma^2 - 4\gamma$
 $= -2(-38) + 3(10) - 4(-2)$
 $= 76 + 30 + 8 = 114$

(b) $P(x) = x^3 + 2x^2 - 3x + 4$.

(i) Now Roots $\alpha+2, \beta+2, \gamma+2$

Then $Y = x+2 \Rightarrow x = Y-2$.

Now $(Y-2)^3 + 2(Y-2)^2 - 3(Y-2) + 4 = 0$
 $Y^3 - 6Y^2 + 12Y - 8 + 2Y^2 - 8Y + 8 - 3Y + 6 + 4 = 0$
 $Y^3 - 4Y^2 + Y + 10 = 0$

$\therefore P(x) = x^3 - 4x^2 + x + 10$

Q16 (b) (i) $P(x) = x^3 + 2x^2 - 3x + 4$.

Now Roots $\alpha^2, \beta^2, \gamma^2$.

$\therefore Y = x^2 \Rightarrow x = \sqrt{Y}$

$(\sqrt{Y})^3 + 2(\sqrt{Y})^2 - 3(\sqrt{Y}) + 4 = 0$

$Y\sqrt{Y} + 2Y - 3\sqrt{Y} + 4 = 0$

$\sqrt{Y}(Y-3) = -2Y-4$

$Y(Y-3)^2 = (2Y+4)^2$

$Y(Y^2 - 6Y + 9) = 4Y^2 + 16Y + 16$

$Y^3 - 6Y^2 + 9Y = 4Y^2 + 16Y + 16$

$Y^3 - 10Y^2 - 7Y - 16 = 0$

$\therefore P(x) = x^3 - 10x^2 - 7x - 16$

(iii) Roots: $\frac{\alpha\beta}{\gamma}, \frac{\alpha\gamma}{\beta}, \frac{\beta\gamma}{\alpha}$ Now $\alpha\beta\gamma = -4$

or $\frac{\alpha\beta\gamma}{\gamma^2}, \frac{\alpha\gamma\beta}{\beta^2}, \frac{\alpha\beta\gamma}{\alpha^2}$

or $-\frac{4}{\alpha^2}, -\frac{4}{\beta^2}, -\frac{4}{\gamma^2}$

from (ii) $-\frac{4}{x}, -\frac{4}{x}, -\frac{4}{x} \therefore Y = -\frac{4}{x}$
 $x = -\frac{4}{Y}$

$\therefore P(Y) = \left(-\frac{4}{Y}\right)^3 - 10\left(-\frac{4}{Y}\right)^2 - 7\left(-\frac{4}{Y}\right) - 16$

$= \frac{-64}{Y^3} - \frac{160}{Y^2} + \frac{28}{Y} - 16$

$0 = -64 - 160Y + 28Y^2 - 16Y^3$

$\therefore 4Y^3 - 7Y^2 + 40Y + 16 = 0$

$\therefore P(x) = 4x^3 - 7x^2 + 40x + 16$